

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2024
Software Engineering (CS-226)
Assignment # 2

Semester: 4th

Student's Name: Muhammad Faisal

Batch: 2020F

Roll#:2022F-BCS-152

Section: C

Total Marks: 3

Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 1	Describe various software engineering processes and activities.	PLO_2 (Academic Education)	C2 (Understanding)

Instructions:

- 1. Use of Generative AI is not allowed.**
- 2. Any answer that is more than 40% identical to another assignment's answer will get 0 marks.**

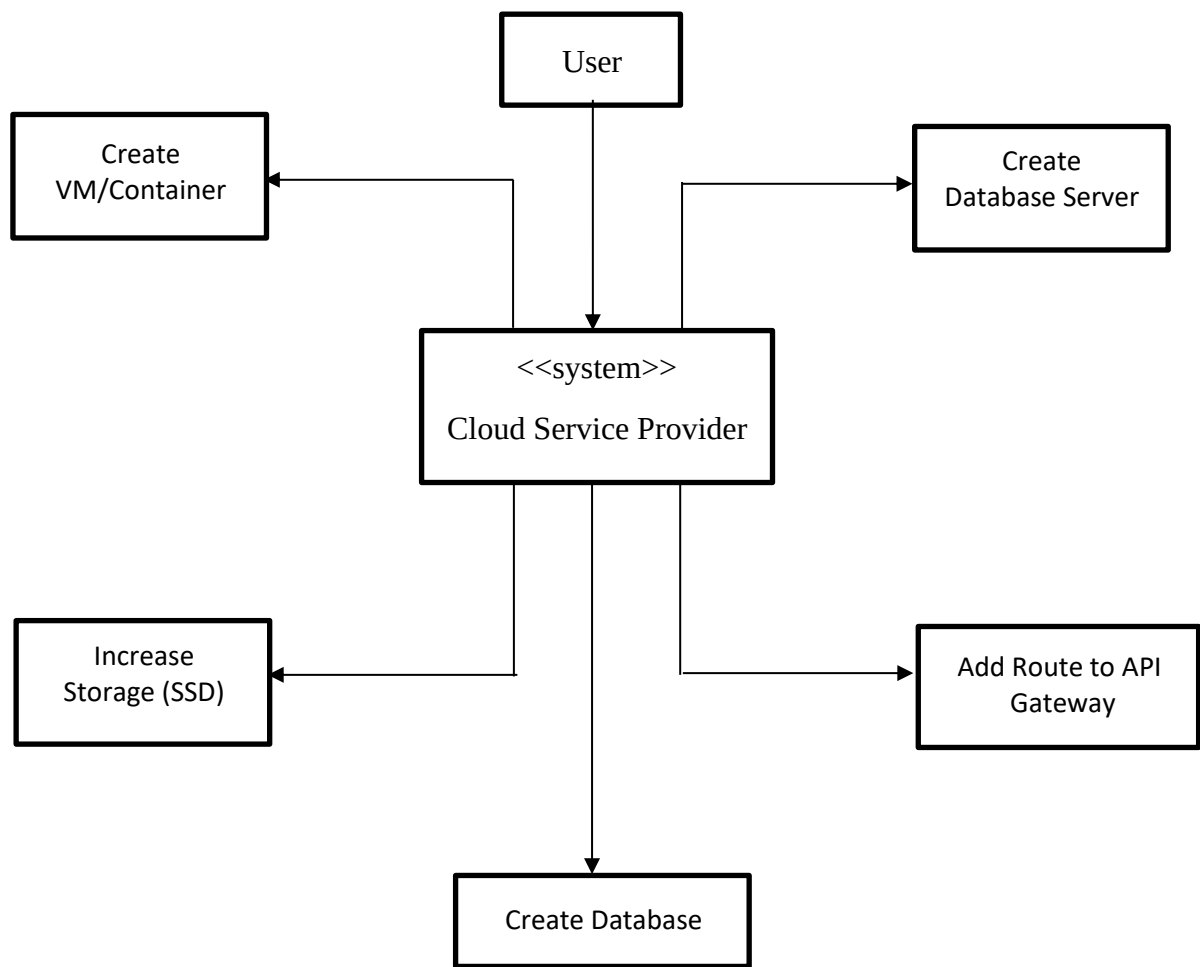
Q1: In one of our last lectures, we learned about the architecture of an application. We also learned about services that a cloud service provides (AWS) which we can use to develop a robust architecture for our app. However, we would also want to learn about how these services are managed because all we mostly see is just a button to click and a loader animation.

Assume that we want to establish our own cloud services provider. However, we want to start from basic. Each user is allowed to use only one VM/Container and only 1 Database Server. Each application hosted by us gets an API gateway, which can be used to configure the routes (urls) for the application.

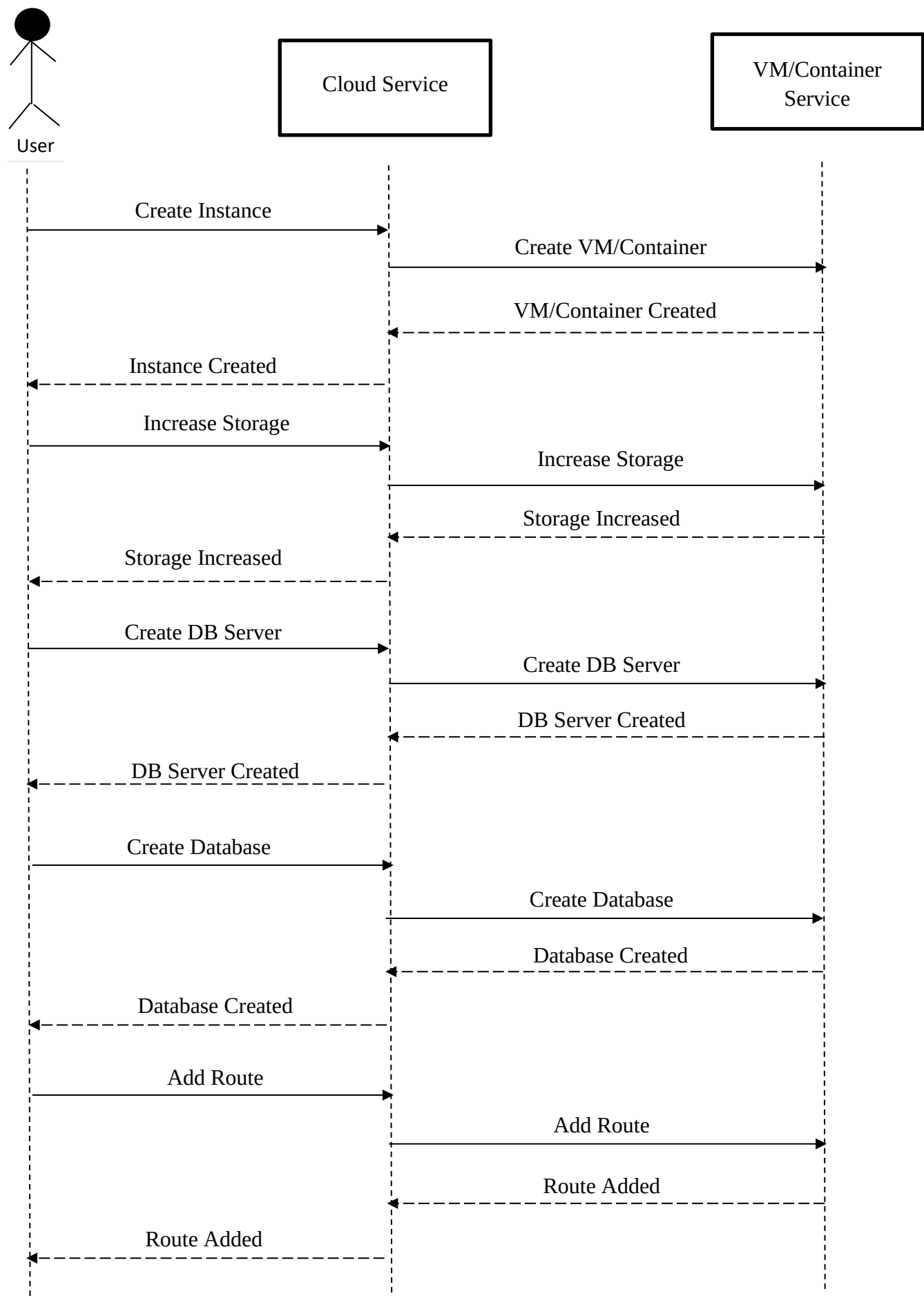
Draw the context diagram for the app as well as Interaction Diagram, Class Diagram and State Diagram (where required) for the following operations.

- 1) An instance(VM/Container) is created.**
- 2) An instance's Storage(SSD) is increased.**
- 3) A database server is created.**
- 4) A database is created within the database server.**
- 5) A route is added to the API gateway.**

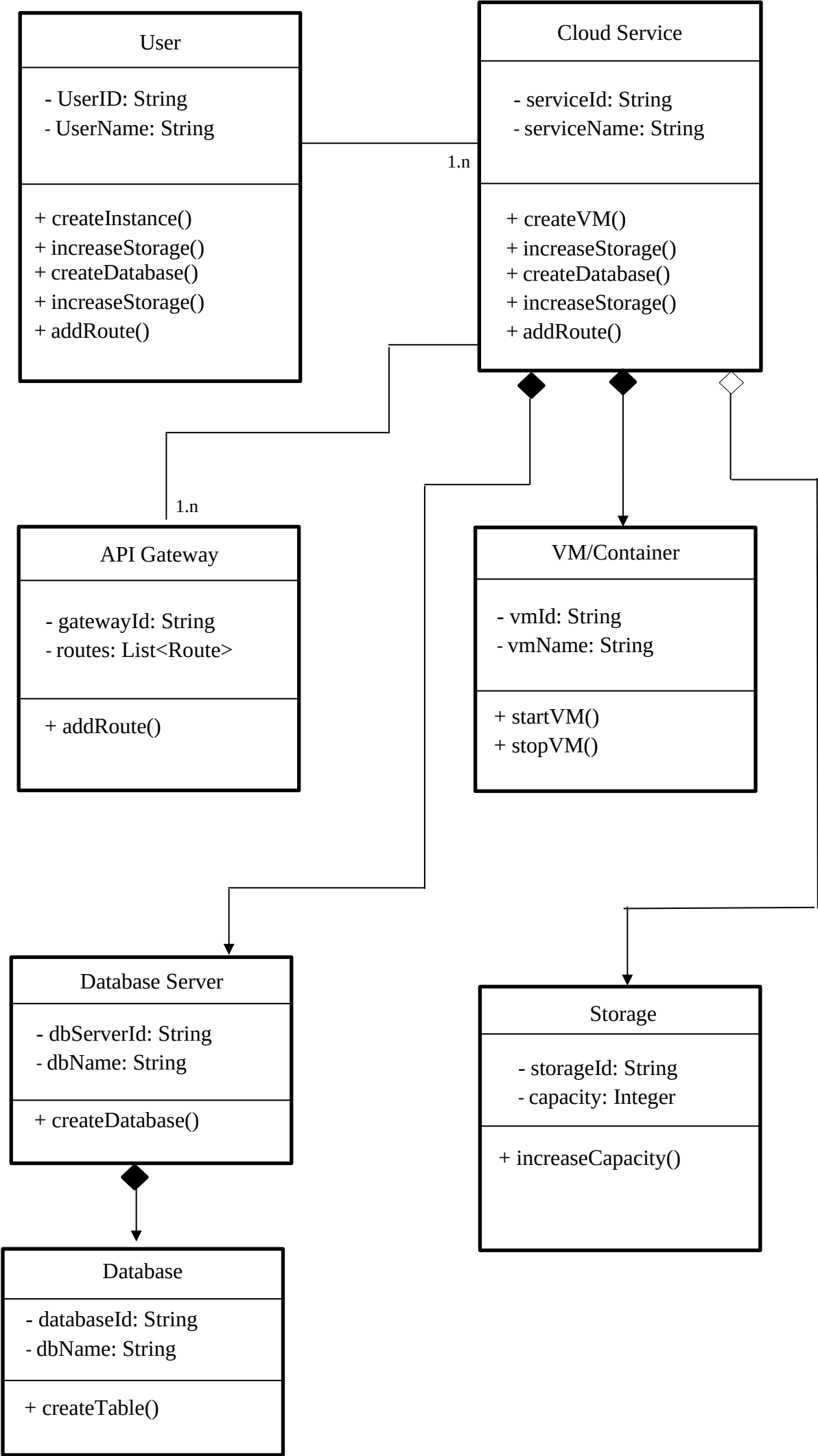
Context Diagram:



Interaction Diagram (Sequence Diagram) :



Class Diagram:



State Diagram

